```
import pandas as pd

# Load orders dataset
orders = pd.read_csv("/content/olist_orders_dataset.csv")

# Convert timestamp to datetime
orders['order_purchase_timestamp'] = pd.to_datetime(
    orders['order_purchase_timestamp'], errors='coerce'
)

# Drop rows with missing timestamps
orders = orders.dropna(subset=['order_purchase_timestamp'])

# Create daily aggregation (number of orders per day)
sales_ts = (
    orders
    .set_index('order_purchase_timestamp')
    .resample('D')
    .size()
    .reset_index(name='sales')
)

# Save as CSV
sales_ts.to_csv("ecommerce_sales_ts.csv", index=False)

print("ecommerce_sales_ts.csv created successfully!")
print(sales_ts.shape)
print(sales_ts.head(10))
```

```
ecommerce_sales_ts.csv created successfully!
(774, 2)
  order_purchase_timestamp  sales
0               2016-09-04      1
1               2016-09-05      1
2               2016-09-06      0
3               2016-09-07      0
4               2016-09-08      0
5               2016-09-09      0
6               2016-09-10      0
7               2016-09-11      0
8               2016-09-12      0
9               2016-09-13      1
```

```
import pandas as pd

# Load sensor dataset
sensor_df = pd.read_csv("/content/sensor

# Convert timestamp column
sensor_df['timestamp'] = pd.to_datetime(

# Sort by time (VERY IMPORTANT for time
sensor_df = sensor_df.sort_values('times
```

```
# Drop duplicate timestamps if any
sensor_df = sensor_df.drop_duplicates(su

# Optional: Reset index
sensor_df = sensor_df.reset_index(drop=T

# Save clean base version
sensor_df.to_csv("sensor_data.csv", inde

print("sensor_data.csv created successfu
print(sensor_df.shape)
print(sensor_df.head())
```

```
sensor_data.csv created successfully!
(220320, 55)
   Unnamed: 0           timestamp  sensor_00  sensor_01  sensor_02  sensor_03
0           0 2018-04-01 00:00:00   2.465394   47.09201    53.2118  46.310760
1           1 2018-04-01 00:01:00   2.465394   47.09201    53.2118  46.310760
2           2 2018-04-01 00:02:00   2.444734   47.35243    53.2118  46.397570
3           3 2018-04-01 00:03:00   2.460474   47.09201    53.1684  46.397568
4           4 2018-04-01 00:04:00   2.445718   47.13541    53.2118  46.397568

   sensor_04  sensor_05  sensor_06  sensor_07  ...  sensor_43  sensor_44  \
0   634.3750   76.45975   13.41146   16.13136  ...   41.92708  39.641200
1   634.3750   76.45975   13.41146   16.13136  ...   41.92708  39.641200
2   638.8889   73.54598   13.32465   16.03733  ...   41.66666  39.351852
3   628.1250   76.98898   13.31742   16.24711  ...   40.88541  39.062500
4   636.4583   76.58897   13.35359   16.21094  ...   41.40625  38.773150

   sensor_45  sensor_46  sensor_47  sensor_48  sensor_49  sensor_50  \
0   65.68287   50.92593  38.194440   157.9861   67.70834   243.0556
1   65.68287   50.92593  38.194440   157.9861   67.70834   243.0556
2   65.39352   51.21528  38.194443   155.9606   67.12963   241.3194
3   64.81481   51.21528  38.194440   155.9606   66.84028   240.4514
4   65.10416   51.79398  38.773150   158.2755   66.55093   242.1875

   sensor_51  machine_status
0   201.3889          NORMAL
1   201.3889          NORMAL
2   203.7037          NORMAL
3   203.1250          NORMAL
4   201.3889          NORMAL

[5 rows x 55 columns]
```

EASY — Sub-step 1

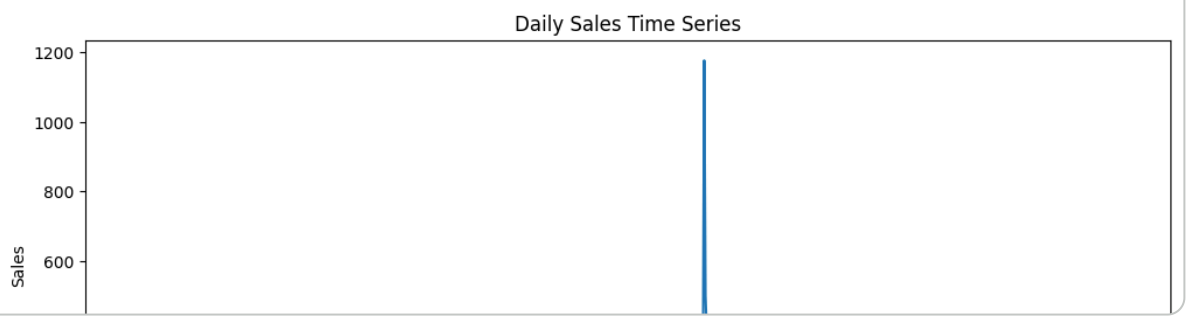
```
import pandas as pd

df = pd.read_csv("ecommerce_sales_ts.csv")
df['order_purchase_timestamp'] = pd.to_datetime(df['order_purchase_timestamp

df = df.sort_values('order_purchase_timestamp')
df.set_index('order_purchase_timestamp', inplace=True)
```

```
df.head()
```

order_purchase_timestamp	sales
2016-09-04	1
2016-09-05	1
2016-09-06	0
2016-09-07	0
2016-09-08	0

Next steps: Generate code with df New interactive sheet

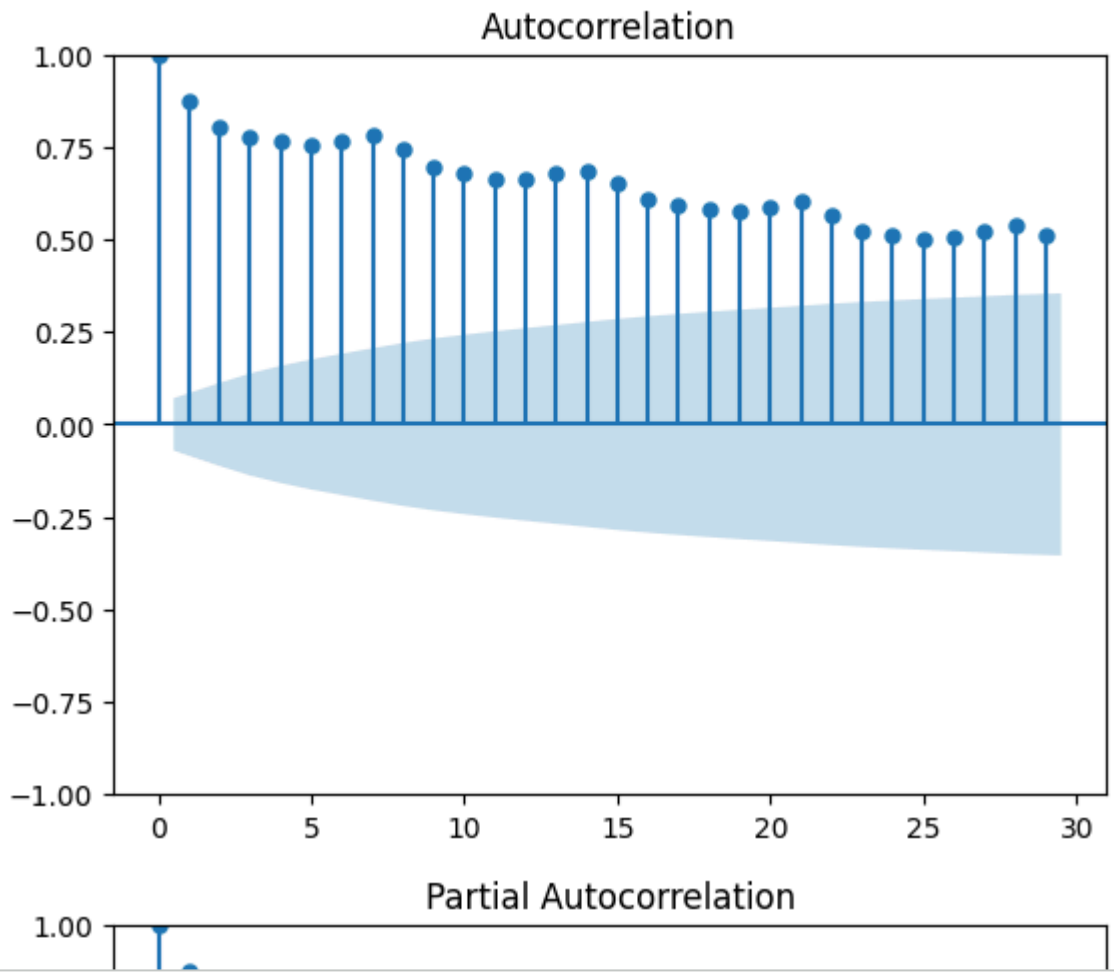
```
import matplotlib.pyplot as plt

plt.figure(figsize=(12,5))
plt.plot(df['sales'])
plt.title("Daily Sales Time Series")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.show()
```




```
from statsmodels.tsa.stattools import adfuller

result = adfuller(df['sales'])

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: -2.3217349227820803
p-value: 0.16505381122077295
```

```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf

plot_acf(df['sales'])
plot_pacf(df['sales'])
plt.show()
```




```
# Missing values
print(df.isnull().sum())

# Duplicates
print(df.index.duplicated().sum())

# Basic stats
print(df.describe())
```

```
sales    0
dtype: int64
0
             sales
count   774.000000
mean    128.476744
std     104.498687
min       0.000000
25%      42.250000
50%     124.000000
75%     198.000000
max    1176.000000
```

EASY — Sub-step 2

```
sensor = pd.read_csv("sensor_data.csv")

sensor['timestamp'] = pd.to_datetime(sensor['timestamp'])
```

```
sensor = sensor.sort_values('timestamp')

sensor.head()
```

	Unnamed: 0	timestamp	sensor_00	sensor_01	sensor_02	sensor_03	sensor_04
0	0	2018-04-01 00:00:00	2.465394	47.09201	53.2118	46.310760	634.3750
1	1	2018-04-01 00:01:00	2.465394	47.09201	53.2118	46.310760	634.3750
2	2	2018-04-01 00:02:00	2.444734	47.35243	53.2118	46.397570	638.8889
3	3	2018-04-01 00:03:00	2.460474	47.09201	53.1684	46.397568	628.1250
4	4	2018-04-01 00:04:00	2.445718	47.13541	53.2118	46.397568	636.4583

5 rows × 55 columns

```
# Missing values
print(sensor.isnull().sum())

# Duplicate timestamps
print(sensor['timestamp'].duplicated().sum())

# Check time gaps
print(sensor['timestamp'].diff().value_counts().head())
```

```
sensor_00          10208
sensor_01            369
sensor_02             19
sensor_03             19
sensor_04             19
sensor_05             19
sensor_06           4798
sensor_07           5451
sensor_08           5107
sensor_09           4595
sensor_10             19
sensor_11             19
sensor_12             19
sensor_13             19
sensor_14             21
sensor_15         220320
sensor_16             31
```

```
sensor_19             16
sensor_20             16
sensor_21             16
sensor_22             41
sensor_23             16
sensor_24             16
sensor_25             36
sensor_26             20
sensor_27             16
sensor_28             16
sensor_29             72
sensor_30            261
sensor_31             16
sensor_32             68
sensor_33             16
sensor_34             16
sensor_35             16
sensor_36             16
sensor_37             16
sensor_38             27
sensor_39             27
sensor_40             27
sensor_41             27
sensor_42             27
sensor_43             27
sensor_44             27
sensor_45             27
sensor_46             27
sensor_47             27
sensor_48             27
sensor_49             27
sensor_50          77017
sensor_51          15383
machine_status         0
dtype: int64
0
timestamp
0 days 00:01:00    220319
Name: count, dtype: int64
```

```
# Check if 'timestamp' is a column before setting it as index
if 'timestamp' in sensor.columns:
    sensor.set_index('timestamp', inplace=True)

# Handle missing values
sensor = sensor.fillna(method='ffill')  # forward fill

# Optional: resample to fixed interval (example: hourly)
sensor = sensor.resample('H').mean(numeric_only=True)
sensor = sensor.fillna(method='ffill')

sensor.head()
```

```
/tmp/ipykernel_2815/1433698865.py:6: FutureWarning: DataFrame.fillna with 'me
  sensor = sensor.fillna(method='ffill')  # forward fill
/tmp/ipykernel_2815/1433698865.py:9: FutureWarning: 'H' is deprecated and wil
  sensor = sensor.resample('H').mean(numeric_only=True)
/tmp/ipykernel_2815/1433698865.py:10: FutureWarning: DataFrame.fillna with 'm
  sensor = sensor.fillna(method='ffill')
```

timestamp	Unnamed: 0	sensor_00	sensor_01	sensor_02	sensor_03	sensor_04	s
2018-04-01 00:00:00	29.5	2.454506	48.556132	53.142359	46.225404	634.120408	
2018-04-01 01:00:00	89.5	2.454244	48.015769	53.106913	45.935328	634.236105	
2018-04-01 02:00:00	149.5	2.453621	48.993053	53.010704	45.593169	634.427072	
2018-04-01 03:00:00	209.5	2.454572	48.654513	52.975985	45.218459	633.416988	
2018-04-01 04:00:00	269.5	2.453785	48.326822	52.868200	45.130930	634.986487	

5 rows × 53 columns

```
sensor.to_csv("sensor_data_cleaned.csv")
```

MEDIUM SUB-STEP 3 (SALES FORECASTING)

```
train_size = int(len(df) * 0.8)

train = df.iloc[:train_size]
test = df.iloc[train_size:]
```

```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(train['sales'], order=(1,1,1))  # baseline choice
model_fit = model.fit()

print(model_fit.summary())
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473
  self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473
  self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473
```

```
  self._init_dates(dates, freq)
                               SARIMAX Results
==============================================================================
Dep. Variable:                  sales   No. Observations:                   61
Model:                 ARIMA(1, 1, 1)   Log Likelihood               -3277.28
Date:                Tue, 14 Apr 2026   AIC                           6560.56
Time:                        15:12:37   BIC                           6573.84
Sample:                    09-04-2016   HQIC                          6565.72
                         - 05-15-2018
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975
------------------------------------------------------------------------------
ar.L1          0.3980      0.034     11.555      0.000       0.330       0.46
ma.L1         -0.8869      0.028    -31.396      0.000      -0.942      -0.83
sigma2      2361.0574     17.140    137.751      0.000    2327.463    2394.65
===================================================================================
Ljung-Box (L1) (Q):                   0.17   Jarque-Bera (JB):              1304
Prob(Q):                              0.68   Prob(JB):
Heteroskedasticity (H):              40.55   Skew:
Prob(H) (two-sided):                  0.00   Kurtosis:
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (comple
```

```
predictions = model_fit.forecast(steps=len(test))
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

mae = mean_absolute_error(test['sales'], predictions)
rmse = np.sqrt(mean_squared_error(test['sales'], predictions))

print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAE: 151.53066613172354
RMSE: 182.86420941286724
```

MEDIUM SUB-STEP 4 (IMPROVED MODEL)

```
from statsmodels.tsa.statespace.sarimax import SARIMAX

model_sarima = SARIMAX(
    train['sales'],
    order=(1,1,1),
    seasonal_order=(1,1,1,7)  # weekly seasonality
)

model_sarima_fit = model_sarima.fit()
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473
  self._init_dates(dates, freq)
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473
  self._init_dates(dates, freq)
```

```
pred_sarima = model_sarima_fit.forecast(steps=len(test))
```

```
mae_sarima = mean_absolute_error(test['sales'], pred_sarima)

print("ARIMA MAE:", mae)
print("SARIMA MAE:", mae_sarima)
```

```
ARIMA MAE: 151.53066613172354
SARIMA MAE: 172.7342405146688
```

MEDIUM SUB-STEP 5 (SENSOR FAILURE PREDICTION)

```
sensor = pd.read_csv("sensor_data.csv")
sensor['timestamp'] = pd.to_datetime(sensor['timestamp'])

# Drop columns that are entirely NaN (e.g., sensor_15)
sensor = sensor.dropna(axis=1, how='all')

# Handle NaNs before calculating rolling features using ffill
sensor = sensor.fillna(method='ffill')

# Example features
sensor['rolling_mean'] = sensor['sensor_01'].rolling(5).mean()
sensor['rolling_std'] = sensor['sensor_01'].rolling(5).std()

# Drop NaNs introduced by rolling calculations (first few rows)
sensor = sensor.dropna(subset=['rolling_mean', 'rolling_std'])
```

```
/tmp/ipykernel_2815/3858292148.py:8: FutureWarning: DataFrame.fillna with 'me
  sensor = sensor.fillna(method='ffill')
```

```
sensor['failure_next_24h'] = sensor['mac

sensor['failure_next_24h'] = sensor['fai
```

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

X = sensor[['rolling_mean', 'rolling_std']]
y = sensor['failure_next_24h']

# Time-based split
split = int(len(X) * 0.8)

X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]
```

```
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier()
```

```
from sklearn.metrics import classification_report

y_pred = model.predict(X_test)

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     44040
           1       0.00      0.00      0.00        24

    accuracy                           1.00     44064
   macro avg       0.50      0.50      0.50     44064
weighted avg       1.00      1.00      1.00     44064
```

HARD SUB-STEP 6 (Rule vs ML Model)

```
THRESHOLD = sensor['sensor_01'].mean() + 2 * sensor['sensor_01'].std()

sensor['rule_pred'] = (sensor['sensor_01'] > THRESHOLD).astype(int)
```

```
sensor['model_pred'] = model.predict(X)
```

```
# Example costs
COST_FN = 1000   # missing a failure (very costly)
COST_FP = 100    # false alarm (less costly)
```

```
def compute_cost(y_true, y_pred):
    cost = 0
    for yt, yp in zip(y_true, y_pred):
        if yt == 1 and yp == 0:
            cost += COST_FN
        elif yt == 0 and yp == 1:
            cost += COST_FP
    return cost
```

```
y_true = y

rule_cost = compute_cost(y_true, sensor['rule_pred'])
model_cost = compute_cost(y_true, sensor['model_pred'])
```

```
print("Rule Cost:", rule_cost)
print("Model Cost:", model_cost)
```

```
Rule Cost: 14742400
Model Cost: 2063600
```

HARD SUB-STEP 7 (Cost Optimization at Scale)

```
NUM_SENSORS = 100000

# Normalize cost per sample
cost_per_sample = model_cost / len(y_tru

daily_cost = cost_per_sample * NUM_SENSC

print("Estimated Daily Cost:", daily_cos
```

```
Estimated Daily Cost: 936654.623359175
```

```
import numpy as np

probs = model.predict_proba(X)[:,1]

thresholds = np.linspace(0, 1, 50)

best_cost = float('inf')
best_threshold = 0

for t in thresholds:
    preds = (probs > t).astype(int)
    cost = compute_cost(y_true, preds)

    if cost < best_cost:
        best_cost = cost
        best_threshold = t

print("Best Threshold:", best_threshold)
print("Minimum Cost:", best_cost)
```

```
Best Threshold: 0.1020408163265306
Minimum Cost: 1566800
```

```
from sklearn.metrics import f1_score

best_f1 = 0
best_f1_threshold = 0

for t in thresholds:
    preds = (probs > t).astype(int)
    f1 = f1_score(y_true, preds)

    if f1 > best_f1:
        best_f1 = f1
```

```
            best_f1_threshold = t

print("Best F1 Threshold:", best_f1_thre
```

```
Best F1 Threshold: 0.4693877551020408
```

Start coding or generate with AI.